

Atomic measures or finite compacta: a complete classification of Iwanik's condition (1)

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Abstract

Choi, Kim, Lee, and Martín ask for which compact Hausdorff spaces K every bounded operator $T : L_1(\mu) \rightarrow C(K)$ has the property that $s \mapsto T^*\delta_s$ is continuous from K to $L_\infty(\mu)$ in the topology of convergence in measure. For each fixed finite measure μ we give a complete dichotomy. If μ is purely atomic, every compact Hausdorff K works. If μ has a nonatomic part of positive measure, exactly the finite spaces K work. The negative direction combines a weak-star-null Rademacher sequence with the fact that every infinite compact Hausdorff space has an infinite compact metrizable continuous image.

Status: full answer, likely valid, pending human review. No prior statement of the classification was found in the bounded novelty search described below.

1 Exact source question

Let μ be a finite measure and K a compact Hausdorff space. For $T \in \mathcal{L}(L_1(\mu), C(K))$, the source writes

$$F_T(s) = T^*(\delta_s) \in L_\infty(\mu) \quad (s \in K).$$

Its condition (1) is that F_T be continuous when $L_\infty(\mu)$ is given the topology of convergence in measure. The definition and its local context appear on printed/PDF page 15 of arXiv:1303.6078:

and

$$\|T - S\|_\infty \leq \|h - h_0\|_\infty + \|h_0 - h_1\|_\infty < \varepsilon^8 + \varepsilon < 2\varepsilon. \quad \square$$

5. THE BISHOP-PHELPS-BOLLOBÁS PROPERTY FOR SOME OPERATORS FROM $L_1(\mu)$ INTO $C(K)$

Throughout this section, we consider only a finite measure μ on a measurable space (Ω, Σ) and **real** Banach spaces $L_1(\mu)$ and $C(K)$. Our aim is to obtain the Bishop-Phelps-Bollobás property for some classes of operators from $L_1(\mu)$ to $C(K)$, sharpening the results about denseness of norm-attaining operators given by Iwanik in 1982 [26].

We use the following standard representation of operators into $C(K)$ [23, Theorem 1 in p. 490].

Lemma 5.1. *Given a bounded linear operator $T : X \rightarrow C(K)$, define $F : K \rightarrow X^*$ by $F(s) = T^*(\delta_s)$, where δ_s is the point measure at $s \in K$. Then, for $x \in X$, the relation $Tx(s) = \langle x, F(s) \rangle$ defines an isometric isomorphism of $\mathcal{L}(X, C(K))$ onto the space of weak* continuous functions from K to X^* with the supremum norm. Moreover, compact operators correspond to norm continuous functions.*

Iwanik [26] considered operators $T \in \mathcal{L}(L_1(\mu), C(K))$ satisfying one of the following conditions:

- (1) The map $s \mapsto T^*\delta_s$ is continuous in measure.
- (2) There exists a co-meager set $G \subset K$ such that $\{T^*\delta_s : s \in G\}$ is norm separable in $L_\infty(\mu)$.

We recall that a subset A is said to be a co-meager subset of K if the set $K \setminus A$ is meager, that is, of first category.

Immediately after Theorem 5.2, on printed/PDF page 16, the authors ask:

“For which topological compact Hausdorff spaces K all operators in $\mathcal{L}(L_1(\mu), C(K))$ satisfy condition (1)?”

The exact source passage is reproduced here:

$$= 2 \int_{\Omega \setminus C} |f(x)| d\mu < \varepsilon$$

On the other hand, we see that $Sg(s_0) = \langle S^*\delta_{s_0}, g \rangle = 1$ because $S^*\delta_{s_0}(\omega) = \text{sign}(f(x)) = \text{sign}(g(\omega))$ for every $\omega \in C$. This completes the proof. $\square$

We do not know, and it is clearly of interest, for which topological compact Hausdorff spaces K all operators in $\mathcal{L}(L_1(\mu), C(K))$ satisfy condition (1).

We recall that a bounded linear operator T from $L_1(\mu)$ into a Banach space X is said to be *Bochner representable* if there is a bounded strongly measurable function $g : \Omega \rightarrow X$ such that

$$Tf = \int f(\omega)g(\omega) d\mu(\omega) \quad (f \in L_1(\mu)).$$

We answer the question for each fixed finite measure μ . “Purely atomic” always means modulo null sets. This includes the zero measure.

2 Proof intuition

The map F_T is automatically weak-star continuous: $s \mapsto \delta_s$ is weak-star continuous in $C(K)^*$, and T^* is weak-star to weak-star continuous. The question is therefore when every bounded weak-star continuous field $F : K \rightarrow L_\infty(\mu)$ must also be continuous in measure.

For a purely atomic measure, weak-star continuity gives scalar continuity on each atom. Only finitely many atoms are needed to cover all but an arbitrarily small measure, so finitely many scalar estimates imply continuity in measure.

On a nonatomic set, by contrast, the Rademacher functions are weak-star null in L_∞ but remain a fixed positive distance from zero in measure. Every infinite compact Hausdorff space maps continuously onto an infinite compact metric space, and such a space contains a nontrivial convergent sequence. Continuous scalar bumps around that sequence let us install the Rademacher functions as the values of one weak-star continuous field. The field, and hence the corresponding operator, fails condition (1).

3 Three lemmas

Lemma 1 (Operator-field correspondence). *For a bounded map $F : K \rightarrow L_\infty(\mu)$, the following are equivalent:*

1. F is weak-star continuous;
2. there is a bounded operator $T : L_1(\mu) \rightarrow C(K)$ satisfying $T^*\delta_s = F(s)$ for every $s \in K$.

In this case one may take

$$(Tf)(s) = \int fF(s) d\mu.$$

Proof. Weak-star continuity makes the displayed scalar function continuous for each $f \in L_1(\mu)$, and boundedness of F gives a bounded operator. The identity for its adjoint follows directly from the definition. Conversely, for a given T , each pairing $\langle F_T(s), f \rangle = (Tf)(s)$ is continuous, which is precisely weak-star continuity. $\square$

Lemma 2 (An infinite metrizable image). *Every infinite compact Hausdorff space K has an infinite compact metrizable continuous image.*

Proof. For $m \geq 1$, let

$$\mathcal{C}_m = \{f \in C(K, \mathbb{R}) : |f(K)| \leq m\}.$$

Each $\mathcal{C}_m$ is closed in the uniform norm: if a limit had $m+1$ distinct values, their positive minimum separation would force all sufficiently close approximants to have at least $m+1$ values.

It also has empty interior. Indeed, if $f \in \mathcal{C}_m$, one of its finitely many level sets is infinite. Choose $m+1$ distinct points in that level set. Since compact Hausdorff spaces are normal, the Tietze extension theorem supplies an arbitrarily small real continuous perturbation taking distinct prescribed values at those points. The perturbed function is not in $\mathcal{C}_m$.

The Banach space $C(K, \mathbb{R})$ is a Baire space, so it is not the union of the closed nowhere-dense sets $\mathcal{C}_m$. Hence some $\pi \in C(K, \mathbb{R})$ has infinite range. Then $M = \pi(K) \subset \mathbb{R}$ is the required infinite compact metrizable continuous image. $\square$

Lemma 3 (A weak-star/measure discontinuity). *Suppose that μ has a nonatomic measurable part E of positive measure. There is a sequence (r_n) in the unit ball of $L_\infty(\mu)$ such that $r_n \rightarrow 0$ weak-star, while*

$$\mu(\{|r_n| > 1/2\}) = \mu(E) > 0 \quad (n \geq 1).$$

Proof. Recursively bisect E into measurable sets of equal measure. On E , let r_n equal 1 or -1 according to which half is selected at the n th level of this dyadic partition, and put $r_n = 0$ off E . The displayed measure identity is immediate.

Let $\mathcal{A}_n$ be the finite sigma-algebra generated by the first n levels, and let $\mathcal{A} = \sigma(\bigcup_n \mathcal{A}_n)$. For $f \in L_1(\mu)$, conditional expectation gives

$$\int_E f r_n d\mu = \int_E \mathbb{E}(f \mid \mathcal{A}) r_n d\mu.$$

The martingale convergence theorem approximates $\mathbb{E}(f \mid \mathcal{A})$ in L_1 by $\mathbb{E}(f \mid \mathcal{A}_N)$. For $n > N$, the latter has integral zero against r_n . Since $\|r_n\|_\infty \leq 1$, it follows that $\int f r_n d\mu \rightarrow 0$. Thus $r_n \rightarrow 0$ weak-star. $\square$

4 Complete classification

Theorem 1. *Let μ be a finite measure and K a compact Hausdorff space. Every operator $T : L_1(\mu) \rightarrow C(K)$ satisfies condition (1) of Choi–Kim–Lee–Martín if and only if*

$$\mu \text{ is purely atomic} \quad \text{or} \quad K \text{ is finite.} \quad (1)$$

Equivalently, if μ has a nonatomic part, the admissible compacta are exactly the finite ones; if μ is purely atomic, every compactum is admissible.

Proof. First suppose that K is finite. It is discrete, so every map from K is continuous and condition (1) is automatic.

Now suppose that μ is purely atomic. A finite purely atomic measure has at most countably many positive-measure atoms A_j , and these cover the space modulo a null set. Every element of $L_\infty(\mu)$ is essentially constant on each A_j . If $F = F_T$ and $c_j(s)$ denotes its value on A_j , then

$$c_j(s) = \frac{1}{\mu(A_j)} \langle F(s), \mathbf{1}_{A_j} \rangle.$$

Consequently every c_j is continuous by weak-star continuity of F .

Fix $s_0 \in K$ and positive a, ε . Choose finitely many atoms $A_1, \dots, A_N$ whose complement has measure less than ε . For s in a sufficiently small neighborhood of s_0 , $|c_j(s) - c_j(s_0)| \leq a$ for $1 \leq j \leq N$. Therefore

$$\mu(\{|F(s) - F(s_0)| > a\}) < \varepsilon.$$

This is continuity in measure.

It remains to prove necessity when μ has a nonatomic part and K is infinite. By Lemma 2, choose a continuous surjection $\pi : K \rightarrow M$ onto an infinite compact metric space. There are distinct points $t_n, t_0 \in M$ with $t_n \rightarrow t_0$. Passing to a subsequence, choose pairwise disjoint open sets U_n containing t_n , whose closures avoid t_0 and whose union accumulates only at t_0 . By normality, choose $\varphi_n \in C(M, [0, 1])$ with

$$\varphi_n(t_n) = 1, \quad \text{supp } \varphi_n \subset U_n.$$

Take the sequence (r_n) from Lemma 3 and define

$$F_0(t_0) = 0, \quad F_0(t) = \sum_{n=1}^{\infty} \varphi_n(t) r_n \quad (t \in M).$$

At most one term is nonzero, so $\|F_0(t)\|_\infty \leq 1$. Away from t_0 the field is locally either zero or a scalar multiple of one fixed r_n , and is therefore norm-continuous. At t_0 , fix $f \in L_1(\mu)$. Since $\langle r_n, f \rangle \rightarrow 0$, all sufficiently late bumps have uniformly small pairing with f ; a neighborhood of t_0

misses the finitely many early supports. Thus $\langle F_0(t), f \rangle \rightarrow 0$ as $t \rightarrow t_0$. Hence F_0 is weak-star continuous on M .

Set $F = F_0 \circ \pi$. Lemma 1 produces $T : L_1(\mu) \rightarrow C(K)$ with $F_T = F$. Choose $s_n \in K$ with $\pi(s_n) = t_n$. Compactness gives a convergent subnet $s_{n_\alpha} \rightarrow s_0$, and continuity of π forces $\pi(s_0) = t_0$. But

$$F(s_{n_\alpha}) = r_{n_\alpha}, \quad F(s_0) = 0,$$

and Lemma 3 shows that this subnet does not converge in measure. Thus F_T is not continuous in measure. This constructs a counterexample whenever both alternatives in (1) fail. $\square$

5 Verification, scope, and novelty audit

The construction is qualitative but every logical interface can be checked directly:

1. Lemma 1 verifies that no representability assumption on the weak-star continuous field is missing.
2. The Baire argument in Lemma 2 avoids any false sequentiality claim about a general compact Hausdorff K ; only its metrizable image is used to choose a sequence.
3. Lemma 3 proves weak-star nullity for every $f \in L_1(\mu)$, not merely for dyadic simple functions.
4. A convergent subnet of the chosen lifts transfers the failure of measure convergence back from M to K .

The theorem concerns the fixed finite measure appearing in the source question. It does not assert a uniform statement over all measures, nor does it decide when condition (1) is equivalent to the Bishop–Phelps–Bollobás property in settings outside the hypotheses of the source paper.

The bounded novelty search checked the run’s solution, attempt, proof-gap, and ledger indexes; the parsed local arXiv corpus; exact arXiv-id and title queries; the exact question; and close searches combining “Iwanik”, “condition (1)”, weak-star continuity, convergence in measure, atomic measures, and compact Hausdorff spaces. It also checked later citations and closely related Bishop–Phelps–Bollobás papers returned by those searches. No later paper stating this dichotomy was found. That is evidence of novelty, not an exhaustive literature guarantee.

Human review should prioritize Lemma 2 and weak-star continuity of F_0 at t_0 . No computer algebra or numerical experiment is used.

References

- [1] Y. S. Choi, S. K. Kim, H. J. Lee, and M. Martín, *The Bishop–Phelps–Bollobás theorem for operators on $L_1(\mu)$* , Journal of Functional Analysis **267** (2014), 214–242; arXiv:1303.6078; doi:10.1016/j.jfa.2014.04.008.
- [2] V. I. Bogachev, *Measure Theory*, Vol. I, Springer, 2007.
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